Jing-Zhang Mobility Commons

An Enterprise–Resident Mobility Operating System

Accept one complete trip before expanding shared feeders. Rail and bus remain the backbone; walking, wheelchair, human, phone and paper routes remain available.

11.41 km²	**3**	**7**	**HOLD**
PROVISIONAL AREA	KEY AREAS	COMPLETE-TRIP LINKS	DEFAULT DECISION

v2.4 candidate | provisional boundary | field authorization 0 | field observations 0 | default decision HOLD

Six Taskbook Outputs | Cases, Identity, Ecosystem, Components, Signage, Calendar

Six taskbook outputs ready for handoff

HOLD / NOT RUN

Methods transfer; claims do not. All designs are candidates; field observations = 0; decision = HOLD.

01 Identity | two loops + switch

Public path stays | service withdraws | handoff is visible

02 Evidence handoff across six roles

- **Public steward** — authorize / HOLD
- **Rail + bus** — capacity / fallback
- **Residents + disabled users** — barrier / refusal
- **Enterprise + operator** — cost / duty
- **Maintenance + data** — repair / handoff
- **Independent review** — pass / repair / withdraw

03 Six case cards | transfer mechanisms, not performance

Punggol ODP
→ shared district interface
× no sensor counts / authority

Seoul S-Map
→ compare scenarios in public
× no coverage / monitoring powers

Helsinki Lab
→ city need to bounded street trial
× trial is not procurement

Woven City
→ human-centred iteration loop
× no private-campus governance

Complete Trip
→ accept the whole trip across roles
× no funding / outcomes

DART GoLink
→ phase first/last mile with operators
× no ridership / cost transfer

04 Components | daily use before icon

1. **Zhongzhiyuan | Failure Archive**
   Xuezhiyuan screening anchor
   staffed arrival + de-identified stop/repair record
2. **AI Origin | Right-to-Arrive Gauge**
   Xueyuanqiao screening anchor
   show broken link, receiver and review date
3. **Dazhongsi | Mobility Timetable**
   station is external to rough PROV-KEY-003
   compare rail, bus, walking, staffed and curb states

05 Signage + annual gates

- Rail time: update / service window / next review
- Switch arrows: automated / staffed / transit paths
- Public receipt: receiver / stop / closure evidence
- Bilingual high contrast + pictogram / tactile / audio

Q1	winter / missed link
Q2	access / community
Q3	rain / heat / curb
Q4	public decision

Annual-week gate: at least 1 authorized real whole-trip audit + published repair/withdrawal receipt

PUBLIC BASE → CANDIDATE SERVICE → HUMAN HANDOFF → HOLD / REPAIR / WITHDRAW

v2.4 candidate · 2026-08-21

v2.4 candidate | provisional boundary | field authorization 0 | field observations 0 | default decision HOLD

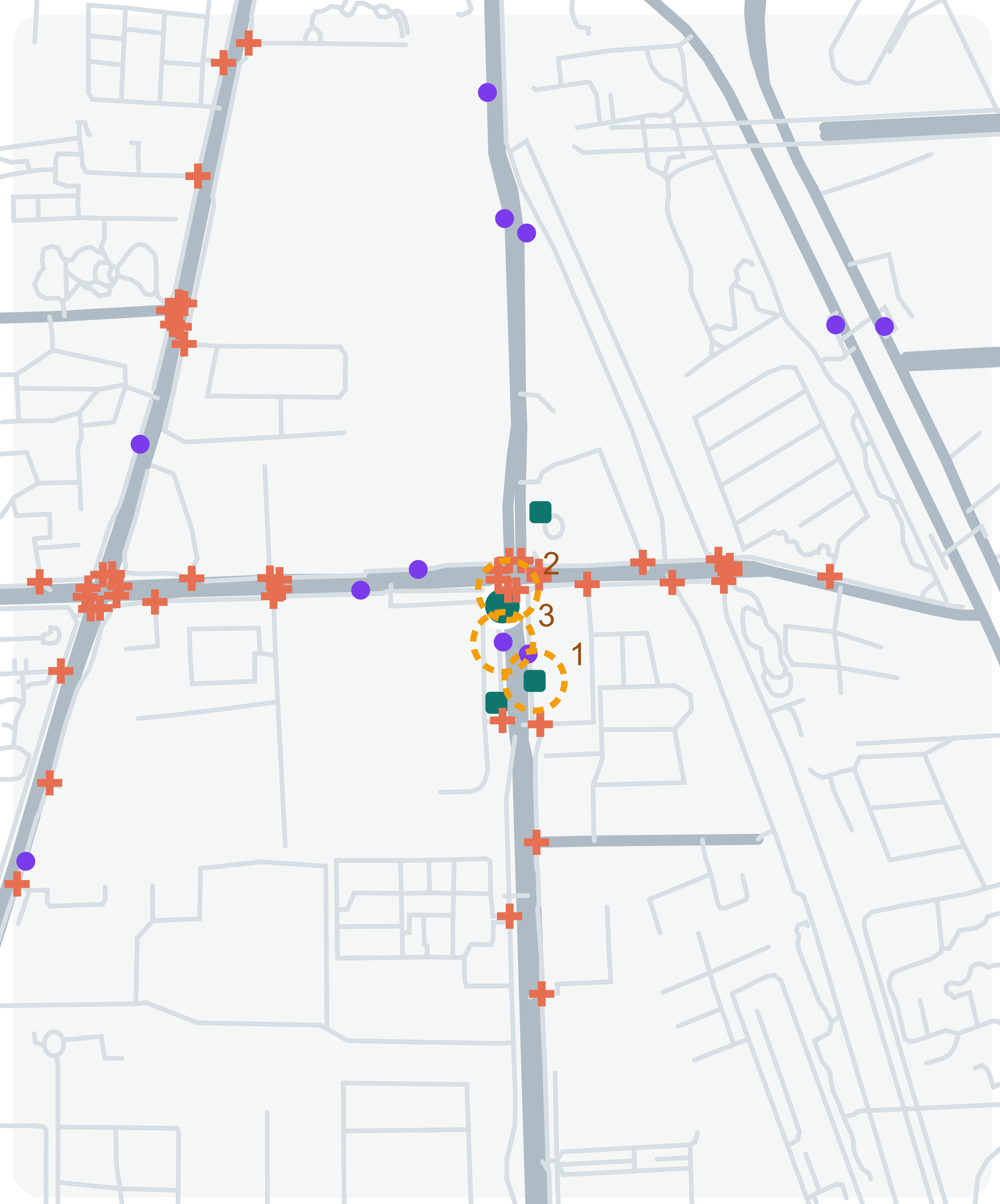
Three Public-Map Interfaces | Entrance, Crossing and Curb Audit Windows

Three Mapped Mobility Anchors

Repair spatial evidence before service release. OSM is background screening only; amber rings mark P0 audit windows, not approved works.

Zhongzhiyuan | Xuezhiyuan arrival interface

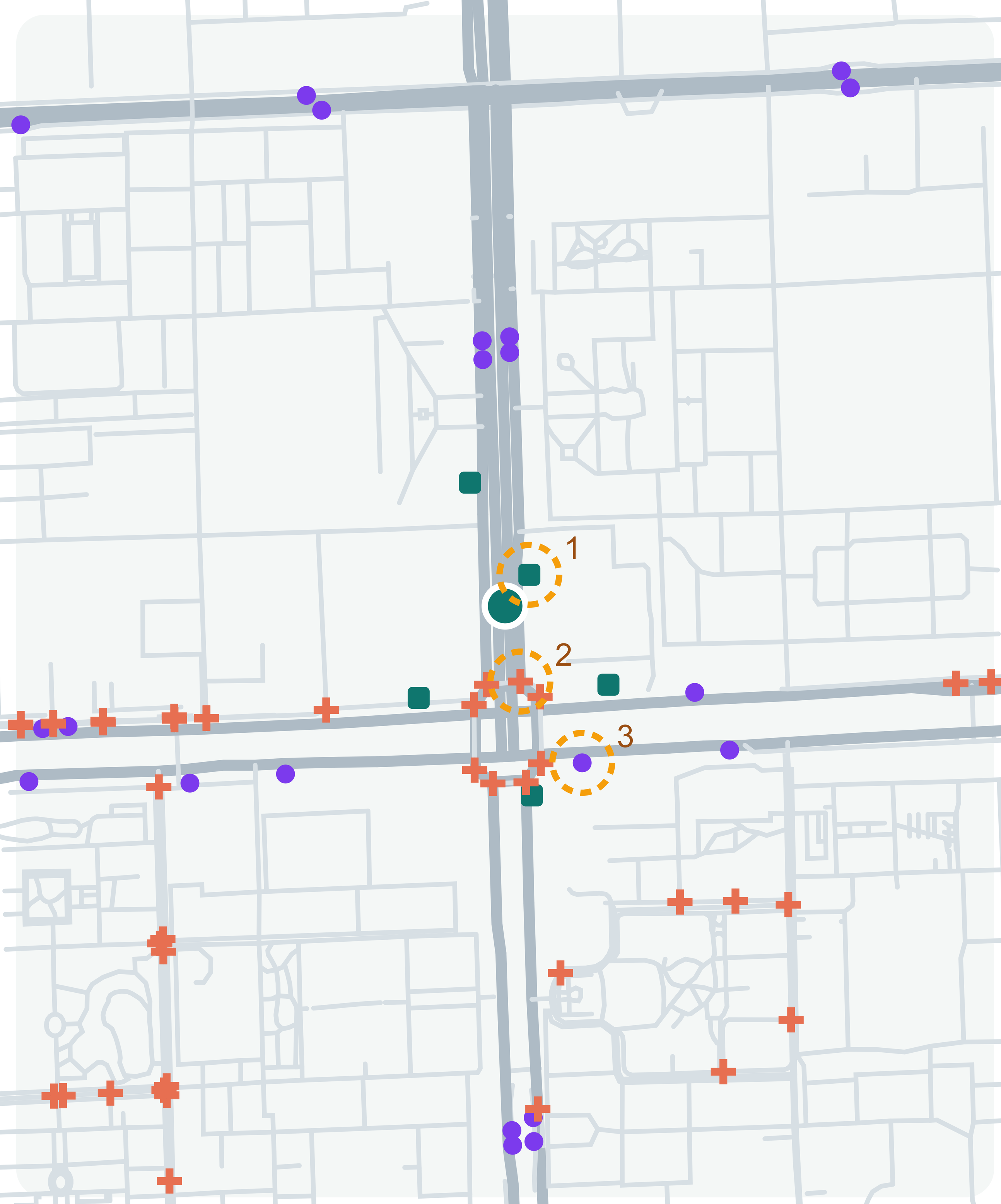
Audit entrances, crossings, shuttle waiting and protected clearance first



1 Station/entrance: Xuezhiyuan
2 Crossing audit window: OSM-mapped, not field-verified
3 Transit/curb audit window: P0 task only

AI Origin | Xueyuanqiao daily-access interface

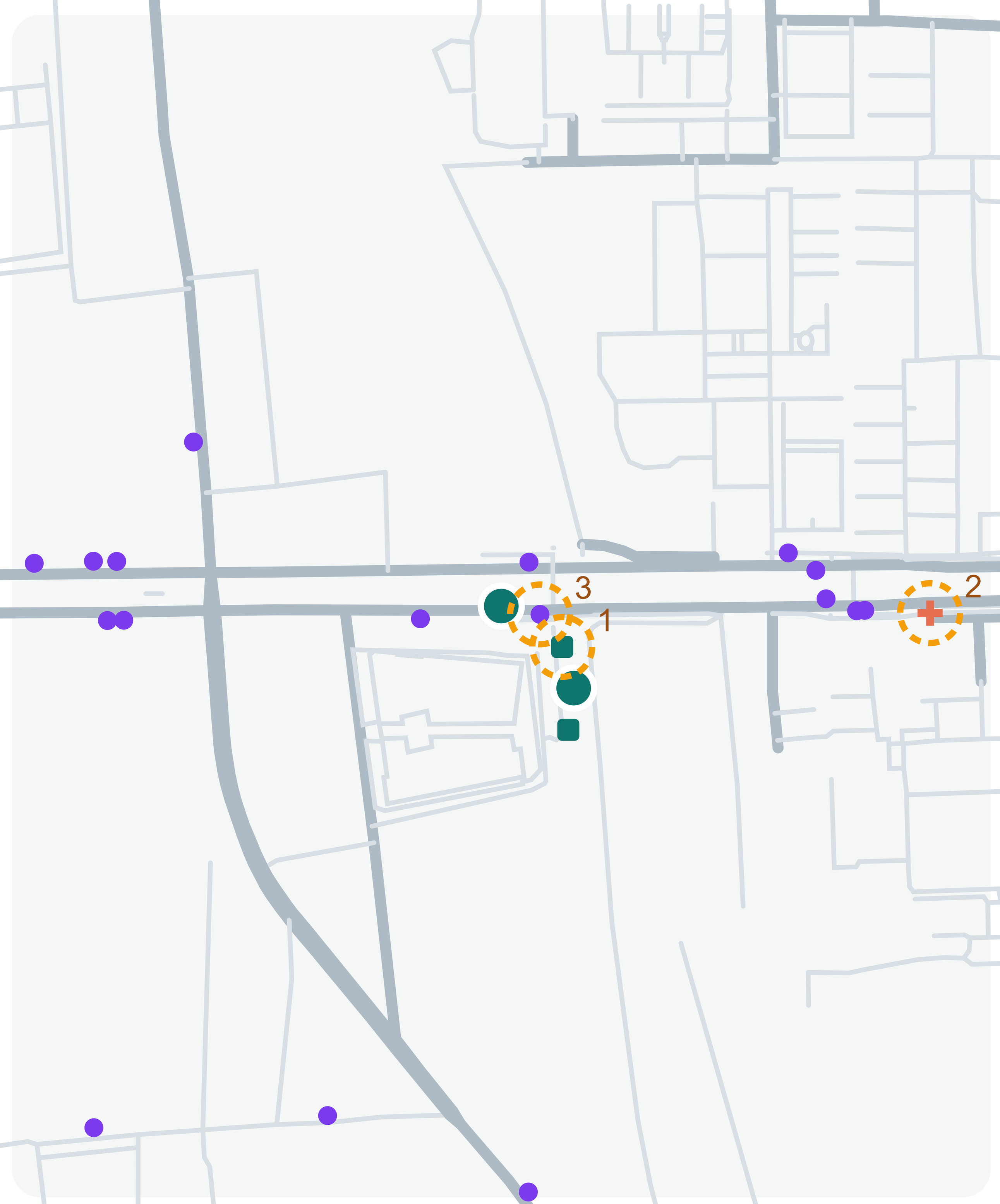
Audit ordinary routes, accessible crossings, staffed service and rain fallback first



1 Station/entrance: Xueyuanqiao
2 Crossing audit window: OSM-mapped, not field-verified
3 Transit/curb audit window: P0 task only

Dazhongsi | Mapped public-transfer anchor

Mapped station is outside the current box; repair geometry before auditing transfer details



1 Station/entrance: Dazhongsi
2 Crossing audit window: OSM-mapped, not field-verified
3 Transit/curb audit window: P0 task only

Geometry conflict: current PROV-KEY-003 does not cover mapped Dazhongsi station.
Until authoritative geometry arrives, the station is an external transfer anchor; the rough rectangle is not a detailed-design boundary.
OpenStreetMap contributors · ODbL 1.0 · accessed 2026-08-21 · field authorization 0 · observations 0 · decision HOLD

v2.4 candidate | provisional boundary | field authorization 0 | field observations 0 | default decision HOLD

OPEN CITY / HAIDIAN / 147228

Three Spatial Interface Prototypes | Entry, Staffed Handover and Withdrawal

MOBILITY COMMONS / 147228

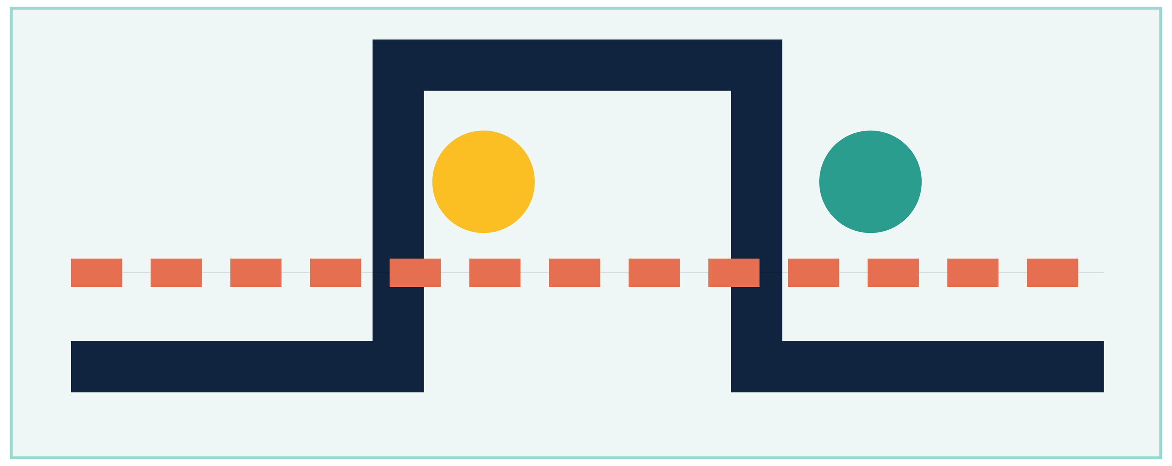
Three spatial interfaces: 1:500 concept → 1:50 handoff detail

DESIGN ONLY

HOLD · 0 authorization · 0 field observations

Shared rule: ordinary or staffed routes come first; each interface has a receiver, dated evidence and a return path; no field dimensions or construction claim.

P-ZHONGZHIYUAN-FORECOURT
Zhongzhiyuan | Arrival + Loading Forecourt

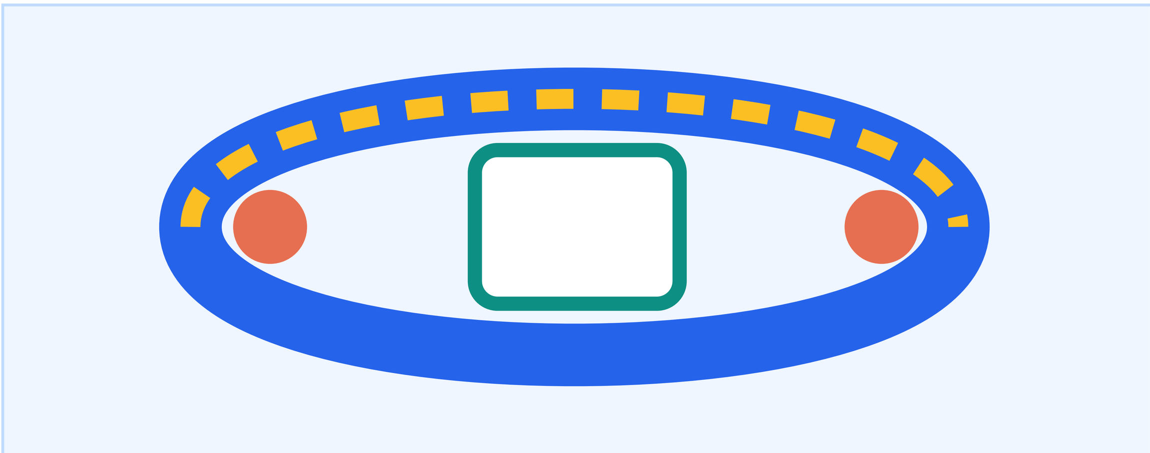


Public route · staffed desk · curb window
Loading stays at HOLD

AI explains grouped time-window conflicts only
1:500 entry, curb and fire-access relation
1:50 staffed takeover and clearance receipt

STOP: no ordinary equivalent
or no clear right-of-way / safety owner

P-AIORIGIN-CARELOOP
AI Origin | Care + Human-Continuity Loop

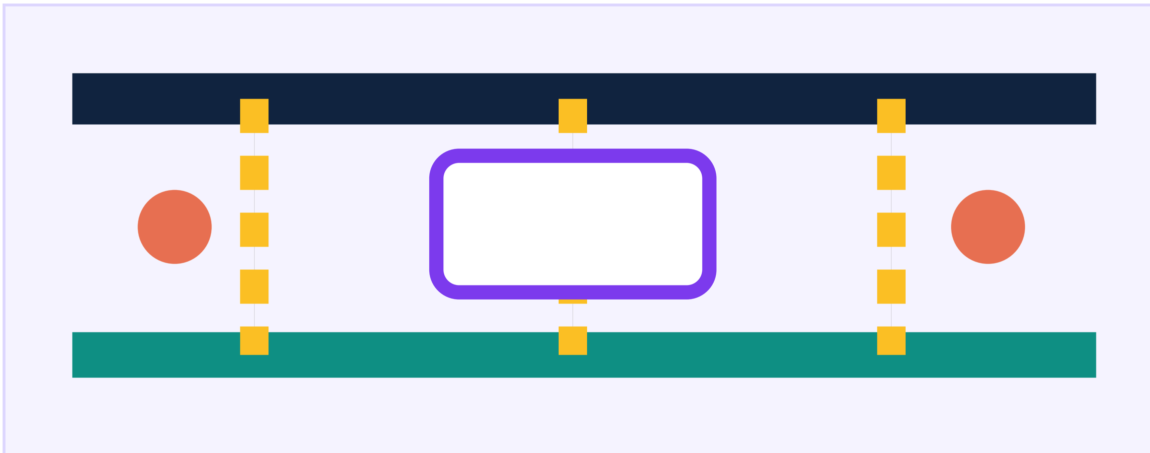


Accessible line · service desk · care pause
Appeal writes back to the receipt

AI groups windows; access never needs an app
1:500 daily route and human-node continuity
1:50 ramp, service desk and carer handoff

STOP: no human equivalent or access breaks
or the privacy boundary is unclear

P-DAZHONGSI-TRANSFERPORCH
Dazhongsi | Rail-to-Curb Transfer Porch



Rail arrival · crossing · bike transfer
Event-day service can pause

AI explains conflict; separation stays reversible
1:500 station, crossing and curb interface
1:50 staffed wayfinding and clearance receipt

STOP: public route is displaced
or crossing / curb evidence is missing

SHARED REVIEW GATES

G1 ordinary route first · G2 receiver and maintenance role · G3 dated observation and public receipt · G4 stop and return to staffed/public transport

These are design references, not a current map; official boundary, dimensions, ownership, permits, field performance and construction conclusions await professional review.

v2.4 candidate | provisional boundary | field authorization 0 | field observations 0 | default decision HOLD

System Spatial Options | Four Alternatives, Five Scales and Five Public Rights

MOBILITY COMMONS / 147228

System spatial options: protect the ground baseline first

CONCEPT ONLY

HOLD · 0 authorization · 0 field observations

Decision order: hard gates → ordinary/accessibility route → most exposed group → reversible handoff; missing local evidence returns to HOLD.

S0 / REJECT
Unmanaged peak

Cars, loading and feeders share one curb

[x] ordinary route unprotected

[x] no human receiver

[x] exposed groups not read back

Return to P0

inventory rights, curb and public route

S1 / ADVANCE TO DESIGN REVIEW
Ground-first multimodal

Rail + bus + walking/accessibility

[ok] ordinary service is visible first

[ok] each node has a receiver

[ok] curb states pause and withdraw

Design review only

not authorization, construction or operation

S2 / REVISE
Air-first feeder

Ground baseline before experiment

[?] airspace, insurance, weather unknown

[?] ground fallback must remain

[?] public interchange cannot be displaced

Return to S1

air layer remains conditional

S3 / REVISE
Extreme-weather fallback

Default layer for rain, outage and repair

[ok] rail/bus/human service stays online

[?] route and roster need readback

[?] paper/telephone withdrawal notice

Return to S1

resilience layer around ground service

Five scales: the same decision from corridor to handoff; no construction dimensions

1	2	3	4	5
1:5000	1:2000	1:500	1:100	1:50
corridor choice	key-area network	node relation	protected zone	handoff detail

FIVE PUBLIC RIGHTS

R1 ordinary access | R2 accessibility and care | R3 rail/bus protection | R4 appeal and pause | R5 privacy and exit

S1 earns only entry to professional design review; official boundary, ownership, right-of-way, permission and performance remain unconfirmed.

Any negative evidence freezes the feeder, keeps the ordinary route and returns to P0; this is an auditable concept comparison, not an operating or ranking claim.

v2.4 candidate | provisional boundary | field authorization 0 | field observations 0 | default decision HOLD

Land-use Shares Computed from GeoJSON / Three Areas, Two Wings, One Responsibility Chain

LAND USE / GEOJSON READBACK

DESIGN PROPOSAL · NOT STATUTORY

Land-use structure | GeoJSON computation

One denominator, four design polygons, reproducible. Not statutory planning control.

Denominator: provisional working area 11,412,825.386 m²

23.4347% | 22.6880% | 29.4943% | 24.3830%

AREA TOTAL
11.413 km²

LU-001 · AI R&D and innovation
23.4347% 2,674,562.492 m²
agent_generated_design · medium confidence

LU-002 · Parks and open space
22.6880% 2,589,339.239 m²
agent_generated_design · medium confidence

LU-003 · Industry, service and commerce
29.4943% 3,366,138.181 m²
agent_generated_design · medium confidence

LU-004 · Community services
24.3830% 2,782,792.823 m²
agent_generated_design · medium confidence

Evidence chain `geometry/land_use.geojson → geometry/site_boundary.geojson → metrics.json`

Shares = declared polygon area / provisional area, displayed to 4 decimals. Recalculate the whole package after a boundary or layer revision.

GeoJSON-derived shares; concept design only; recalculate after any boundary revision.

Three key areas / Three operating roles

MULTIMODAL / v1.1
MOBILITY COMMONS / v1.0

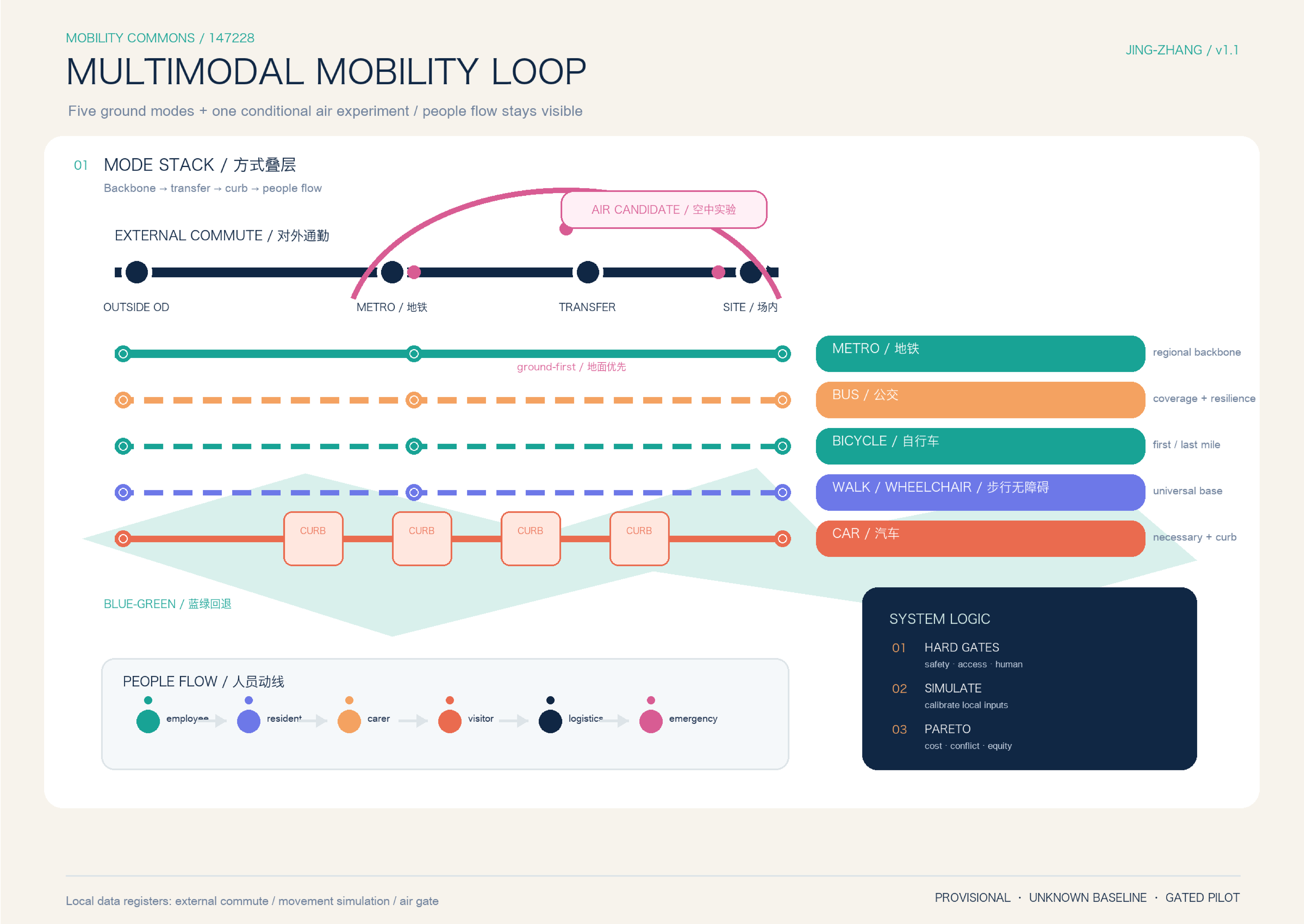
Each area has a minimum pilot, accountable owner and stop condition

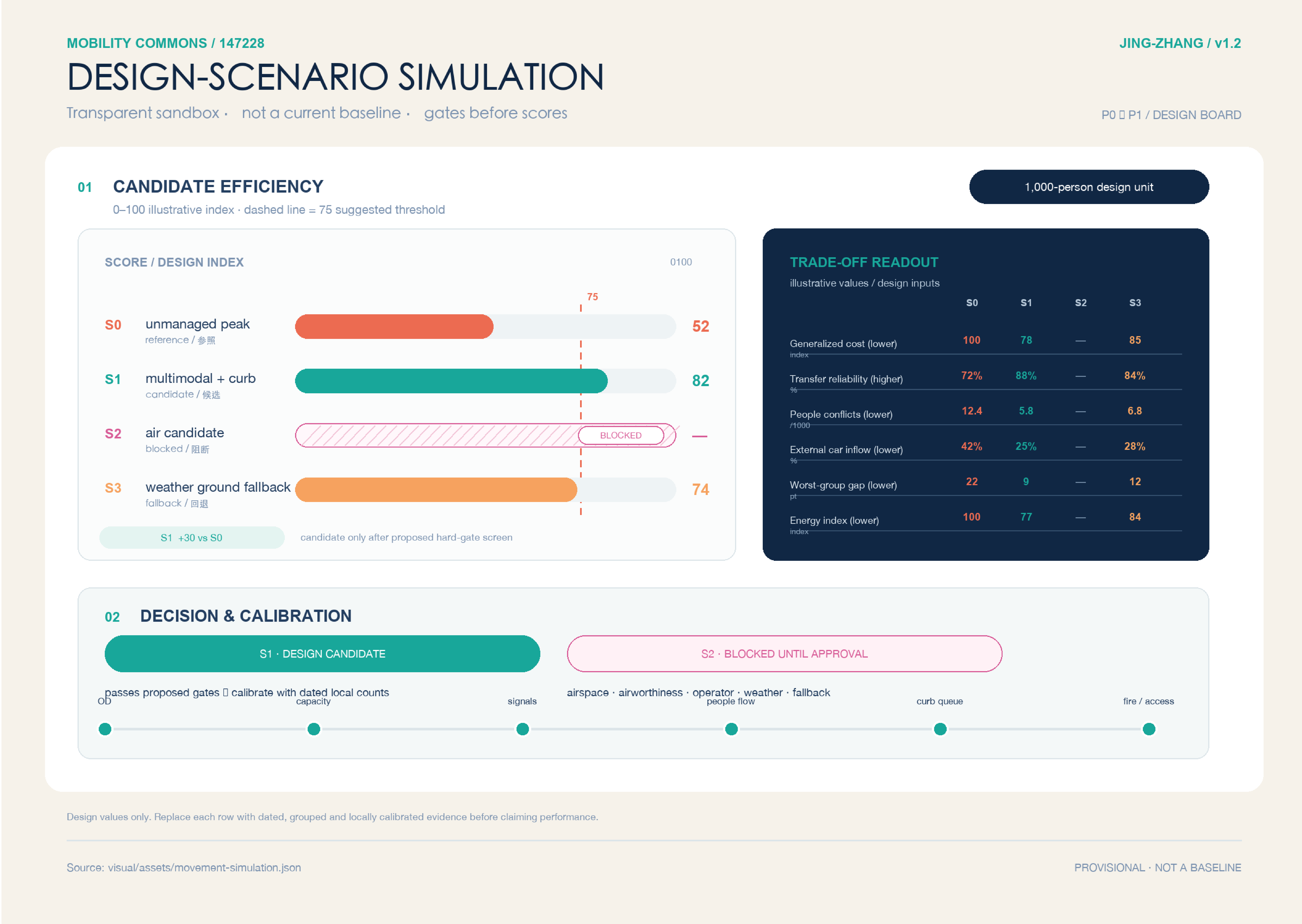
	Area	Role	Stop
01	Zhongzhiyuan enterprise arrival	shuttle merge / loading pilot / reversible	STOP No permanent community ban
02	AI Origin resident daily	human parity / accessible pilot / reversible	STOP No home trajectory
03	Dazhongsi rail + curb	station / cycle / event pilot / reversible	STOP No fire/access blockage

服务水平 / SERVICE LEVELS — 到达连续 route continuity — 换乘可靠 transfer reliability — 路缘有序 curb order — 申诉闭环 complaint closure

PROVISIONAL DESIGN • known / unknown / pilot target kept separate   147228

Five Ground Modes and One Conditional Experiment / Transparent Sandbox for Sensitivity Comparison Only





	S0	S1	S2	S3
Generalized cost (lower) index	100	78	—	85
Transfer reliability (higher) %	72%	88%	—	84%
People conflicts (lower) /1000	12.4	5.8	—	6.8
External car inflow (lower) %	42%	25%	—	28%
Worst-group gap (lower) pt	22	9	—	12
Energy index (lower) index	100	77	—	84

Synthetic sensitivity outputs only; local calibration, field counts and professional review remain missing.

Exit Receipt from the First 168 Hours to Week 12

JING-ZHANG MOBILITY COMMONS | 168 HOURS → 12 WEEKS

Three spatial interfaces · ordinary service first · accountable handoffs · withdrawal before expansion

BOUNDARY Design sequence, not operating authorization; authorization=0 · field observations=0 · baseline=unknown · performance=null · default HOLD

H0 | 0–24h
Make ordinary access work
- BASE: no account or model
- human, phone and paper entry
- name receiver and complaint route
- stop digital layer if missing

Artifact: ordinary route card

H1 | 24–72h
Register space and evidence gaps
- Zhongzhiyuan / AI Origin / Dazhongsi
- route, curb and handoff denominators
- grouped demand, no personal traces
- no receiver/fallback means HOLD

Artifact: route-service atlas

H2 | 72–120h
Rehearse failure offline
- BOOST / BLACKOUT / BEQUEST
- outage, weather, conflict, refusal
- human/public-transport takeover
- negative cases are not outcomes

Artifact: handoff + withdrawal

H3 | 120–168h
Review, then HOLD if needed
- exact tree and evidence map
- maintenance / complaint / withdrawal
- synthetic is never field outcome
- missing evidence: repair or do not release

Artifact: review / archive receipt

THREE SPATIAL INTERFACES

Zhongzhiyuan
Arrival, loading, fire and accessible curb
Refuse: no clearing owner / protected route

AI Origin Community
Daily access, care, wheelchair and human desk
Refuse: digital layer becomes basic access

Dazhongsi
Rail transfer, wayfinding and timed curb
Refuse: capacity / transfer / fire unknown

FIRST 12 WEEKS

W01–02 boundary/consent → W03–04 route audit → W05–07 ordinary parity → W08–09 failure tabletop → W10–11 independent review → W12 HOLD / repair / withdraw

v2.4 candidate | provisional boundary | field authorization 0 | field observations 0 | default decision HOLD

Seven-Link Continuity and Responsibility Transfer

Four Windows, One Ordinary Route

Ordinary service first · AI for aggregation and explanation · authorization 0 · field observations 0 · not_run · HOLD

DO NOT ADVANCE

One door-to-door chain, four service windows that can refuse entry

Each window keeps an ordinary route, a human equivalent, a named receiver and a stop action.

01 AM ARRIVAL
Zhongzhiyuan · weekday arrival

Ordinary: rail/bus + curb window
AI: grouped windows and conflicts
Equivalent: paper + staffed marshal

No receiver/capacity → refuse
Keep public route open

02 DAILY ACCESS
AI Origin · daily services

Ordinary: walk-in/phone/paper desk
AI: grouped windows, no traces
Equivalent: continuous access route

Basic access harmed → disable
Return to P0 route review

03 PM RETURN
Dazhongsi · evening transfer

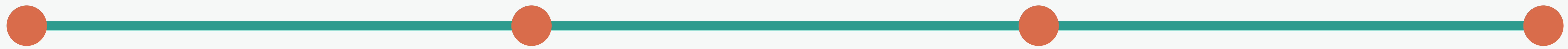
Ordinary: timetable + staffed wayfinding
AI: explain conflict, suggest separation
Equivalent: public crossing information

Capacity unknown → remove feeder
Keep public route open

04 OUTAGE / WEATHER
Whole chain · disruption fallback

Ordinary: human/rail/bus/paper
AI: offline rollback checklist
Equivalent: staffed handoff + public info

No safe handoff → freeze
Keep HOLD



Every handoff leaves the same receipt fields

state · node · service group · ordinary route · AI used or not · human equivalent · receiver · accepted/refused · fallback · version · next decision

Current boundary

4 states · 4 handoff seams · 4 positive controls · 6 negative HOLD fixtures · synthetic review only · no field result

All seven links must pass; one broken link returns the complete trip to HOLD.

Six Responsibility Seats and the Public-Service Floor

Mobility Responsibility Transfer

7 resource units · 8 groups · field HOLD · no real transfer accepted

REAL TRANSFERS 0

Every service must answer three questions

Who receives the burden? What is the non-AI equivalent? What record permits refusal or rollback?

01 Route and curb

Recipient: transport/accessibility review

Denominator: segments, crossings, staffed windows

Equivalent: signs, staff, public transport

Missing record → freeze reservation, return to P0

02 Rail and bus

Recipient: rail/bus operations

Denominator: headways, capacity, transfer windows

Equivalent: timetable and staffed desk

Unknown capacity → remove feeder demand

03 Enterprise booking

Recipient: enterprise mobility coordinator

Denominator: curb minutes and clearing staff

Equivalent: paper register and marshal

Public route displaced → refuse request

04 Resident daily access

Recipient: community and carer observers

Denominator: group attempts and human windows

Equivalent: walk-in, telephone, paper

Basic access harmed → disable recommendation

05 Maintenance and assets

Recipient: field maintenance

Denominator: inspection hours, repairs, assets

Equivalent: manual ticket and removable asset

No maintainer → remove, return to safe state

06 Data and equity

Recipient: data and independent review

Denominator: groups, negatives, valid windows

Equivalent: aggregate paper register

Average hides gaps → hold expansion

Writeback fields

event ID · group · interface · resource · accepted/refused · fallback · responsible role · evidence version · next decision

Six responsibility seats remain unconfirmed; public transport and human-equivalent routes remain available.

Where Review Starts and Where It Must Stop

Known	Provisional area 11,412,825.386 m²; three candidate key areas; green ratio 12.3423%; public-space ratio 7.3281%.
Still null	Enterprise and resident demand, curb occupancy, transfer delay, accessible completion, conflicts, and complaint duration.
Stop conditions	Boundary, demand, safety, operator, accessibility, privacy, or responsibility gaps keep the package at HOLD.
External-method boundary	Complete Trip, DART GoLink, disability journey research, Shanghai wheelchair transit, and FHWA curb guidance provide methods only.
High-water protection	This PDF belongs only to the v2.4 candidate branch. Local PASS is not a maintainer score and cannot replace the protected 69-point version.